

# A CLOSED FORM AND A LUCAS CONGRUENCE FOR APÉRY'S COMPANION SEQUENCE

[AUTHOR]

ABSTRACT. We give a two-term closed form for the numerators  $b_n$  of Apéry's approximations to  $\zeta(3)$ , a Lucas-type congruence  $p^3 b_{ap+r} \equiv b_a a_r \pmod{p}$  for them, and a correction to a published integrality claim. Every theorem and lemma below is proved in Lean 4 and checked by its kernel, and is followed by the name of the declaration that proves it; the one conjecture is labelled as such.

## 1. INTRODUCTION

Apéry's proof that  $\zeta(3) \notin \mathbb{Q}$  rests on the pair of sequences  $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$  and a companion  $b_n$  with  $b_n/a_n \rightarrow \zeta(3)$ . The first sequence is arithmetically well understood: it satisfies a Lucas congruence  $a_{ap+r} \equiv a_a a_r \pmod{p}$  (Gessel [2]), so it respects base- $p$  digits. The companion  $b_n$  has attracted no such statement, and it is usually written with a nested alternating inner sum, which makes it awkward to handle modulo  $p$ .

We report three things.

First, a closed form (Theorem 2.6):  $b_n$  is the same binomial sum as  $a_n$  weighted by a *two-term* harmonic expression,

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}), \quad H_m^{(r)} = \sum_{j=1}^m j^{-r}.$$

The nested sum disappears.

Second, a Lucas-type congruence for the companion itself (Theorem 2.7):

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p} \quad (a, r < p).$$

The factor  $p^3$  is the weight of  $\zeta(3)$ . The same argument gives the analogous congruence for Franel's companion at weight 2 (Theorem 2.8).

Third, a correction. Brown and Zudilin [1] record  $d_n^5 P_n \in \mathbb{Z}$  for the numerators of their cellular approximations to  $\zeta(5)$ , where  $d_n = \text{lcm}(1, \dots, n)$ . Their own printed value  $P_2 = 1190161/384$  does not satisfy this. At  $n = 2$  the defect is exactly  $12 = 2^2 \cdot 3$  (Theorem 3.2); consequently any constant  $c$  for which  $c d_n^5 P_n$  is always integral is divisible by 12. Whether 12 suffices for every  $n$  we state only as Conjecture 3.3: it is supported by computation and by a proof for the primes  $p \geq 5$  given elsewhere, and it is not formalised.

Finally we record a transcription slip in the same author's earlier work (Section 4); it is benign, and we do not formalise it.

Throughout,  $p$  is prime, and for  $x, y \in \mathbb{Q}$  we write  $x \equiv y \pmod{p}$  to mean  $v_p(x-y) \geq 1$ .  
PInt, PDvd, PCong

## 2. APÉRY'S COMPANION

**Definition 2.1.**  $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$  and  $a_n = \sum_{k=0}^n A(n, k)$ . A, apéry

**Lemma 2.2.** For  $r, s < p$ ,  $A(ap+r, bp+s) \equiv \binom{a}{b}^2 \binom{a+b}{b}^2 A(r, s) \pmod{p}$ . A\_digits

This is Lucas' theorem applied to each binomial, together with the fact that a carry  $r+s \geq p$  makes both sides vanish. choose\_digits, choose\_carry\_zero

**Definition 2.3.**  $P(n) = 34n^3 + 51n^2 + 27n + 5$ , and  $b_n$  is defined by

$$b_0 = 0, \quad b_1 = 6, \quad (n+2)^3 b_{n+2} = P(n+1) b_{n+1} - (n+1)^3 b_n.$$

Ppol, bApery

This is Apéry's recurrence;  $b_2 = 351/4$  and  $b_3 = 62531/36$ .

**Definition 2.4.**  $\tilde{b}_n = \sum_{k=0}^n A(n, k) (2H_n^{(3)} - H_k^{(3)})$ .

Harm, bMin

**Lemma 2.5.**  $(m+2)^3 \tilde{b}_{m+2} - P(m+1) \tilde{b}_{m+1} + (m+1)^3 \tilde{b}_m = 0$  for all  $m \geq 0$ . *bMin-rec*

The proof is a single telescoping step: applying the recurrence operator to the summand of  $\tilde{b}$  produces an exact difference  $\Psi(m, k+1) - \Psi(m, k)$ , whose sum collapses to the boundary values  $\Psi(m, 0) = \Psi(m, m+3) = 0$ . The two Zeilberger-type certificates involved are polynomial identities in  $\mathbb{Q}[m, k]$ . *star, star\_pre*

**Theorem 2.6.**  $\tilde{b}_n = b_n$  for all  $n$ ; that is,

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n.$$

*bMin-eq-bApery, apery-b-harmonic-closed-form*

**Theorem 2.7.** For  $a, r < p$ ,  $p^3 b_{ap+r} \equiv b_a a_r \pmod{p}$ .

*bApery-lucas*

The weight 3 enters as follows: modulo  $p$  the summands with  $k \not\equiv 0$  contribute nothing after multiplication by  $p^3$ , while the layer  $k = jp$  reproduces  $H^{(3)}$  at  $\lfloor \cdot/p \rfloor$ , since  $p^3 \sum_{m \leq y, p|m} m^{-3} = \sum_{j \leq y/p} j^{-3}$ . The remaining digit factorisation is Lemma 2.2.

**Theorem 2.8.** Let  $f_n = \sum_k \binom{n}{k}^3$  and  $\beta_n = \sum_k \binom{n}{k}^3 (\frac{1}{4} H_k^{(2)} + \frac{3}{4} (H_k^{(1)})^2 - \frac{3}{4} H_k^{(1)} H_{n-k}^{(1)})$ . Then for  $p \neq 2$  and  $a, r < p$ ,

$$p^2 \beta_{ap+r} \equiv \beta_a f_r \pmod{p}.$$

*franel, bFranel, bFranel-lucas*

### 3. A DENOMINATOR DEFECT

**Definition 3.1.**  $d_n = \text{lcm}(1, \dots, n)$ , and  $P_2 = 1190161/384$  is the value printed in [1]. *dlcm, BZ\_P2*

**Theorem 3.2.**  $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$ , while  $12 d_2^5 P_2 = 1190161 \in \mathbb{Z}$ ; and if  $c \in \mathbb{N}$  satisfies  $c d_2^5 P_2 \in \mathbb{Z}$ , then  $12 \mid c$ . *BZ\_P2-not-integral, BZ\_P2-factor-twelve, BZ-twelve-minimal*

Here  $d_2 = 2$  and  $1190161 \equiv 1 \pmod{12}$ , so at  $n = 2$  the defect is exactly  $2^2 \cdot 3$ . Specialising to  $n = 2$ , any  $c \in \mathbb{N}$  with  $c d_n^5 P_n \in \mathbb{Z}$  for all  $n$  is divisible by 12.

**Conjecture 3.3.**  $12 d_n^5 P_n \in \mathbb{Z}$  for every  $n \geq 1$ .

Theorem 3.2 shows that no smaller constant can serve. Conjecture 3.3 itself is *not* formalised: it is supported by exact computation, and its restriction to the primes  $p \geq 5$  is proved elsewhere, conditionally on two finite identities verified but not machine-checked.

## 4. AN ERRATUM

This section is expository and is not formalised.

Section 8 of [5] proves that one of  $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$  is irrational. Its linear forms are built from a very-well-poised rational function, defined at (8.2) by

$$\tilde{R}(t) = (h_0 + 2t) \frac{\Gamma(h_0 + t)^r \prod_{j=1}^q \Gamma(h_j + t)}{\Gamma(1 + t)^r \prod_{j=1}^q \Gamma(1 + h_0 - h_j + t)},$$

and then normalised arithmetically as  $R = N\tilde{R}$  with  $N = \prod_{j=r+1}^q (h_0 - 2h_j)! / \prod_{j=1}^r (h_j - 1)!^2$ . Equation (8.7), which restates  $R(t)$  as an explicit product, omits the factor  $(h_0 + 2t)$ . The omission is in the published text and in the arXiv source, not in a transcription of it.

The factor is needed. It is the unique factor of  $\tilde{R}$  that is antisymmetric under  $t \mapsto -t - h_0$ , so it is what makes the well-poised symmetry (8.5),

$$\tilde{R}(-t - h_0) = -\tilde{R}(t),$$

hold; and that symmetry is what forces  $B_{jk} = (-1)^j B_{j, h_0 - k}$  for the partial-fraction coefficients, hence the vanishing of the coefficients of the *even* zeta values in  $F(h) = \sum_j A_{j-1} \zeta(j-1) - A_0$ , which is the content of Lemma 19 there.

The effect of the omission is not a loss of precision but a reversal of parity. Taking the paper's own parameters  $r = 3$ ,  $q = 13$ ,  $\eta = (91; 27, 27, 27, 29, 30, \dots, 38)$ ,  $h_0 = 91n + 2$ ,  $h_j = \eta_j n + 1$ , and computing the expansion of  $F$  at  $n = 1$  in exact arithmetic: with  $(h_0 + 2t)$  present, the coefficients of  $\zeta(3)$  and of  $\zeta(4), \zeta(6), \zeta(8), \zeta(10), \zeta(12)$  are all exactly zero and only  $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$  survive, as Theorem 3 of [5] requires; reading (8.7) literally, the even zeta values survive instead and the odd ones vanish.

So (8.7) should read  $R(t) = (h_0 + 2t) \cdot [\text{the printed product}]$ . Equations (8.2), (8.5) and (8.6) are consistent with the factor present, and the results of §8 are unaffected.

## 5. THE DEVELOPMENT

Lean 4 (v4.33.0-rc1) with Mathlib cd580e54; ten modules, 2107 lines; no `sorry` and no `native.decide`. Besides the results above it contains Gessel's congruence  $a_{ap+r} \equiv a_a a_r$  [2] and its analogue for the Brown–Zudilin double sum

$$Q_n = \sum_{k,l} \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n},$$

in both cases with the base- $p$  digit-product form (`apery_lucas`, `Q_lucas`). Every declaration cited above satisfies

```
#print axioms → [propext, Classical.choice, Quot.sound].
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Build with lake exe cache get && lake build.
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## REFERENCES

- [1] F. Brown and W. Zudilin, *On cellular rational approximations to  $\zeta(5)$* , preprint [arXiv:2210.03391v3](#) (2022).
- [2] I. M. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), no. 3, 362–368.
- [3] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381. [arXiv:1910.09336](#).
- [4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28, LNCS 12699, 625–635.
- [5] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.